Theorem 0.1 (Cholesky decomposition): *Let $A_{n\times n}$ be a real, **symmetric**, and **positive definite** matrix. Then A has a unique factorization*

$$A = LL^T,$$

where $L_{n\times n}$ is a lower triangular matrix with strictly positive diagonal entries (i.e., $L_{ii} > 0$).

Explicit formula: The elements of L are computed as follows:

$$L_{ii} = \sqrt{A_{ii} - \sum_{k=1}^{i-1} (L_{ik})^2}, \quad \text{for } i = 1, \ldots, n$$

$$L_{ij} = \frac{1}{L_{jj}} \left(A_{ij} - \sum_{k=1}^{j-1} L_{ik} L_{jk} \right), \quad \text{for } i > j$$